

Month-end value

X_{month} = last available daily value of X
 applied to Nifty, gold (USD), USD/INR and the Liquid index

Gold price in rupees

$$G_{INR,t} = G_{USD,t} \times FX_t$$

G_{USD} = \$ per troy ounce · FX = ₹ per US\$ (FRED DEXINUS)

Liquid index (daily accrual)

$$L_d = L_{d-1} \left(1 + \frac{y_d}{365} \right)$$

y_d = 91-day T-bill yield on day d

Implied T-bill rate (check)

$$y_d = \left(\frac{L_d}{L_{d-1}} - 1 \right) \times 365$$

General monthly return

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

growth of ₹1 over the month · 239 months (Feb 2000 – Dec 2019)

Nifty 50 total return

$$r_{Nifty,t} = \frac{P_t}{P_{t-1}} - 1 + \frac{dy}{12}$$

P_t = month-end Nifty close · dy = 1.3% dividend yield per year

Gold return (Indian investor)

$$r_{Gold,t} = \frac{G_{INR,t}}{G_{INR,t-1}} - 1 = (1 + r_{USD,t}) \frac{FX_t}{FX_{t-1}} - 1$$

r_{USD} = gold's dollar return · FX ratio = rupee depreciation

Liquid return

$$r_{Liquid,t} = \frac{L_t}{L_{t-1}} - 1$$

L_t = index at month end

Correlation

$$\rho_{ij} = \frac{\text{Cov}(r_i, r_j)}{\sigma_i \sigma_j}$$

+1 move together · 0 unrelated · -1 opposite

Engine The SIP engine (shared by all six strategies)

sip/engine.py · each strategy chooses: target w, pro-rata or smart split, and none / calendar / band rebalancing

Instalment

$$C_t = A(1 + g)^{\lfloor t/12 \rfloor}$$

$A = ₹10,000$ · $g = 0$ (no step-up) · $t = 0$ (Feb 2010) ... 118 (Dec 2019)

Split the instalment: pro-rata

$$x_i = w_i C_t$$

x_i = rupees into jar i · w_i = target share

Split the instalment: smart (fill the gaps first)

$$V = \sum_i h_i + C_t \quad G = \sum_i gap_i$$

$$gap_i = \max(w_i V - h_i, 0)$$

$$x_i = gap_i C_t / G \quad \text{if } G \geq C_t$$

$$x_i = gap_i + w_i (C_t - G) \quad \text{if } G < C_t$$

h_i = rupees already in jar i · V = wealth after the instalment ·

gap_i = shortfall of jar i vs target

Buy

$$h_i \leftarrow h_i + x_i$$

" $\leftarrow$ " = becomes

Rebalance: none

never rebalance

jars drift freely

Rebalance: calendar (every 12 months)

$$\text{if } n_{\text{since}} \geq 12: \quad trade_i = w_i \sum_j h_j - h_i$$

$$h_i \leftarrow w_i \sum_j h_j \quad n_{\text{since}} \leftarrow 0$$

n_{since} = months since the last rebalance · $trade_i > 0$ buy, < 0 sell

Rebalance: band (drift > 5%)

$$drift = \max_i \left| \frac{h_i}{\sum_j h_j} - w_i \right|$$

$$\text{if } drift > 0.05: \quad h_i \leftarrow w_i \sum_j h_j$$

drift = the largest gap between a jar's actual share and its target

Trading fee

$$Fee = c \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \left(1 - \frac{Fee}{V_{\text{start}}} \right)$$

$c = 0.001$ (0.1%) · V_{start} = total just before the fee

Market move

$$h_i \leftarrow h_i (1 + r_{i,t})$$

$r_{i,t}$ = return of asset i in month t

Month's return (time-weighted)

$$TWR_t = \frac{V_{\text{end}}}{V_{\text{start}}} - 1$$

V_{end} = total at month end · V_{start} = total after buying/rebalancing, before the fee

Order inside every month

$C_t \rightarrow x_i \rightarrow h_i + x_i \rightarrow \text{rebalance?} \rightarrow Fee \rightarrow \times (1 + r_{i,t}) \rightarrow TWR_t$

repeated for $t = 0 \dots 118$ (119 months); each month starts from last month's jars

S1-S3 Strategies 1-3: fixed targets

01_equity_sip · 02_equal_weight_sip · 03_fixed_60_20_20_sip

Strategy 1 · Equity SIP (100% Nifty)

Target (fixed, every month)

$$w = (1, 0, 0)$$

Nifty / Gold / Liquid

Split the instalment: pro-rata

$$x_i = w_i C_t$$

x_i = rupees into jar i · w_i = target share

Rebalance: none

never rebalance

jars drift freely

Strategy 2 · Equal-weight SIP

Target (fixed, every month)

$$w = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$$

one third each

Split the instalment: pro-rata

$$x_i = w_i C_t$$

x_i = rupees into jar i · w_i = target share

Rebalance: none

never rebalance

jars drift freely

Strategy 3 · 60/20/20 SIP

Target (fixed, every month)

$$w = (0.60, 0.20, 0.20)$$

Nifty / Gold / Liquid

Split the instalment: pro-rata

$$x_i = w_i C_t$$

x_i = rupees into jar i · w_i = target share

Rebalance: calendar

if $n_{since} \geq 12$:

$$h_i \leftarrow w_i \sum_j h_j$$

every 12 months (each January)

Then, for all three, every month

Instalment

$$C_t = A (1 + g)^{\lfloor t/12 \rfloor}$$

$A = ₹10,000$ · $g = 0$ (no step-up) · $t = 0$ (Feb 2010) ... 118 (Dec 2019)

Buy

$$h_i \leftarrow h_i + x_i$$

" $\leftarrow$ " = becomes

Trading fee

$$Fee = c \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \left(1 - \frac{Fee}{V_{start}} \right)$$

$c = 0.001$ (0.1%) · V_{start} = total just before the fee

Market move

$$h_i \leftarrow h_i (1 + r_{i,t})$$

$r_{i,t}$ = return of asset i in month t

Month's return (time-weighted)

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} = total at month end · V_{start} = total after buying/rebalancing, before the fee

S4 Risk-parity SIP

04_risk_parity_sip/strategy.py · target from sip/optimize.py, then the shared engine with smart split and 5% band

Look-back statistics (previous 120 months only)

$$\mu_i = \frac{1}{120} \sum_{k=1}^{120} r_{i,t-k}$$

$$\Sigma_{ij} = \frac{1}{119} \sum_{k=1}^{120} (r_{i,t-k} - \mu_i)(r_{j,t-k} - \mu_j)$$

μ = average monthly returns · Σ = covariance matrix (Σ_{ij} = variance)

Risk parity (equal risk contributions)

$$RC_i = w_i (\Sigma w)_i \quad \overline{RC} = \frac{1}{3} \sum_i RC_i$$

$$w^{RP} = \arg \min_w \sum_i (RC_i - \overline{RC})^2$$

RC_i = asset i 's contribution to portfolio variance ($\Sigma_i RC_i = w^T \Sigma w$)

Constraints (solver: SLSQP)

$$0.10 \leq w_i \leq 0.70 \quad \sum_i w_i = 1$$

every asset between 10% and 70%, fully invested

Walk-forward re-fit (every February)

$$w^{(Feb Y)} = \text{optimiser}(r_{Feb(Y-10)}, \dots, r_{Jan Y})$$

$Y = 2010 \dots 2019$ · target held for 12 months · no look-ahead

Instalment

$$C_t = A(1+g)^{\lfloor t/12 \rfloor}$$

$A = ₹10,000$ · $g = 0$ (no step-up) · $t = 0$ (Feb 2010) ... 118 (Dec 2019)

Split the instalment: smart (fill the gaps first)

$$V = \sum_i h_i + C_t \quad G = \sum_i gap_i$$

$$gap_i = \max(w_i V - h_i, 0)$$

$$x_i = gap_i C_t / G \quad \text{if } G \geq C_t$$

$$x_i = gap_i + w_i (C_t - G) \quad \text{if } G < C_t$$

h_i = rupees already in jar i · V = wealth after the instalment · gap_i = shortfall of jar i vs target

Buy, then market move

$$h_i \leftarrow h_i + x_i$$

$$h_i \leftarrow h_i (1 + r_{i,t})$$

$r_{i,t}$ = return of asset i in month t (fee is taken in between)

Rebalance: band (drift > 5%)

$$drift = \max_i \left| \frac{h_i}{\sum_j h_j} - w_i \right|$$

$$\text{if } drift > 0.05: \quad h_i \leftarrow w_i \sum_j h_j$$

drift = the largest gap between a jar's actual share and its target

Trading fee

$$Fee = c \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \left(1 - \frac{Fee}{V_{start}} \right)$$

$c = 0.001$ (0.1%) · V_{start} = total just before the fee

Month's return (time-weighted)

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} = total at month end · V_{start} = total after buying/rebalancing, before the fee

S5 Max-Sharpe SIP

05_max_sharpe_sip/strategy.py · target from sip/optimize.py, then the shared engine with smart split and 5% band

Look-back statistics (previous 120 months only)

$$\mu_i = \frac{1}{120} \sum_{k=1}^{120} r_{i,t-k}$$

$$\Sigma_{ij} = \frac{1}{119} \sum_{k=1}^{120} (r_{i,t-k} - \mu_i)(r_{j,t-k} - \mu_j)$$

μ = average monthly returns · Σ = covariance matrix (Σ_{ij} = variance)

Max Sharpe (best return per unit of risk)

$$\mu_p = w^\top \mu \quad \sigma_p = \sqrt{w^\top \Sigma w}$$

$$w^{MS} = \arg \max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}$$

μ_p, σ_p = expected monthly return and volatility of the mix (rf = 0)

Constraints (solver: SLSQP)

$$0.10 \leq w_i \leq 0.70 \quad \sum_i w_i = 1$$

every asset between 10% and 70%, fully invested

Walk-forward re-fit (every February)

$$w^{(Feb Y)} = \text{optimiser}(r_{Feb(Y-10)}, \dots, r_{Jan Y})$$

$Y = 2010 \dots 2019$ · target held for 12 months · no look-ahead

Instalment

$$C_t = A(1+g)^{\lfloor t/12 \rfloor}$$

$A = ₹10,000$ · $g = 0$ (no step-up) · $t = 0$ (Feb 2010) ... 118 (Dec 2019)

Split the instalment: smart (fill the gaps first)

$$V = \sum_i h_i + C_t \quad G = \sum_i gap_i$$

$$gap_i = \max(w_i V - h_i, 0)$$

$$x_i = gap_i C_t / G \quad \text{if } G \geq C_t$$

$$x_i = gap_i + w_i (C_t - G) \quad \text{if } G < C_t$$

h_i = rupees already in jar i · V = wealth after the instalment · gap_i = shortfall of jar i vs target

Buy, then market move

$$h_i \leftarrow h_i + x_i$$

$$h_i \leftarrow h_i (1 + r_{i,t})$$

$r_{i,t}$ = return of asset i in month t (fee is taken in between)

Rebalance: band (drift > 5%)

$$drift = \max_i \left| \frac{h_i}{\sum_j h_j} - w_i \right|$$

$$\text{if } drift > 0.05: \quad h_i \leftarrow w_i \sum_j h_j$$

drift = the largest gap between a jar's actual share and its target

Trading fee

$$Fee = c \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \left(1 - \frac{Fee}{V_{start}} \right)$$

$c = 0.001$ (0.1%) · V_{start} = total just before the fee

Month's return (time-weighted)

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} = total at month end · V_{start} = total after buying/rebalancing, before the fee

S6 Optimized SIP (½ RP + ½ MS)

06_optimized_sip/strategy.py · target from sip/optimize.py, then the shared engine with smart split and 5% band

Look-back statistics (previous 120 months only)

$$\mu_i = \frac{1}{120} \sum_{k=1}^{120} r_{i,t-k}$$

$$\Sigma_{ij} = \frac{1}{119} \sum_{k=1}^{120} (r_{i,t-k} - \mu_i)(r_{j,t-k} - \mu_j)$$

μ = average monthly returns · Σ = covariance matrix (Σ_{ij} = variance)

Risk parity (equal risk contributions)

$$RC_i = w_i (\Sigma w)_i \quad \overline{RC} = \frac{1}{3} \sum_i RC_i$$

$$w^{RP} = \arg \min_w \sum_i (RC_i - \overline{RC})^2$$

RC_i = asset i 's contribution to portfolio variance ($\Sigma_i RC_i = w^T \Sigma w$)

Max Sharpe (best return per unit of risk)

$$\mu_p = w^T \mu \quad \sigma_p = \sqrt{w^T \Sigma w}$$

$$w^{MS} = \arg \max_w \frac{w^T \mu}{\sqrt{w^T \Sigma w}}$$

μ_p, σ_p = expected monthly return and volatility of the mix (rf = 0)

Optimized SIP target

$$w_i^{Opt} = \frac{1}{2} w_i^{RP} + \frac{1}{2} w_i^{MS}$$

the average of the risk-parity and max-Sharpe weights, each fitted with the same μ, Σ , limits

Constraints (solver: SLSQP)

$$0.10 \leq w_i \leq 0.70 \quad \sum_i w_i = 1$$

every asset between 10% and 70%, fully invested

Walk-forward re-fit (every February)

$$w^{(Feb Y)} = \text{optimiser}(r_{Feb(Y-10)}, \dots, r_{Jan Y})$$

$Y = 2010 \dots 2019$ · target held for 12 months · no look-ahead

Instalment

$$C_t = A(1+g)^{\lfloor t/12 \rfloor}$$

$A = ₹10,000$ · $g = 0$ (no step-up) · $t = 0$ (Feb 2010) ... 118 (Dec 2019)

Split the instalment: smart (fill the gaps first)

$$V = \sum_i h_i + C_t \quad G = \sum_i gap_i$$

$$gap_i = \max(w_i V - h_i, 0)$$

$$x_i = gap_i C_t / G \quad \text{if } G \geq C_t$$

$$x_i = gap_i + w_i (C_t - G) \quad \text{if } G < C_t$$

h_i = rupees already in jar i · V = wealth after the instalment · gap_i = shortfall of jar i vs target

Buy, then market move

$$h_i \leftarrow h_i + x_i$$

$$h_i \leftarrow h_i (1 + r_{i,t})$$

$r_{i,t}$ = return of asset i in month t (fee is taken in between)

Rebalance: band (drift > 5%)

$$drift = \max_i \left| \frac{h_i}{\sum_j h_j} - w_i \right|$$

$$\text{if } drift > 0.05: \quad h_i \leftarrow w_i \sum_j h_j$$

drift = the largest gap between a jar's actual share and its target

Trading fee

$$Fee = c \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \left(1 - \frac{Fee}{V_{start}} \right)$$

$c = 0.001$ (0.1%) · V_{start} = total just before the fee

Month's return (time-weighted)

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} = total at month end · V_{start} = total after buying/rebalancing, before the fee

Scoring Scoring every strategy

sip/metrics.py · applied to each strategy's 119 monthly results

XIRR (money-weighted return per year)

$$\sum_{k=0}^{N-1} \frac{-C_k}{(1 + XIRR)^{k/12}} + \frac{V_{final}}{(1 + XIRR)^{N/12}} = 0$$

solved for XIRR (brentq) · $N = 119$ instalments · V_{final} = value at Dec 2019

TWR CAGR (strategy's own growth rate)

$$CAGR = \left[\prod_{t=1}^N (1 + TWR_t) \right]^{12/N} - 1$$

Volatility (per year)

$$\sigma = \text{std}(TWR_t) \times \sqrt{12}$$

Sharpe (vs 0%)

$$Sharpe = \frac{12 \times \text{mean}(TWR_t)}{\sigma}$$

risk-free rate = 0

Sharpe vs Liquid

$$Sharpe_{Liq} = \frac{12 \times \text{mean}(TWR_t - r_{Liquid,t})}{\sigma}$$

only the return above the T-bill (Liquid) counts

Sortino

$$DD = \sqrt{\text{mean}(\min(TWR_t, 0)^2)} \times \sqrt{12}$$
$$Sortino = \frac{12 \times \text{mean}(TWR_t)}{DD}$$

only downside moves count as risk

Maximum drawdown

$$G_t = \prod_{k \leq t} (1 + TWR_k)$$
$$MDD = \min_t \left(\frac{G_t}{\max_{k \leq t} G_k} - 1 \right)$$

deepest fall of ₹1 from a previous high

Calmar

$$Calmar = \frac{CAGR}{|MDD|}$$

Worst wealth drop (account value)

$$WWD = \min_t \left(\frac{V_t}{\max_{k \leq t} V_k} - 1 \right)$$

V_t = rupee value of the account (includes new instalments)

Wealth multiple

$$\frac{V_{final}}{\sum_t C_t}$$

Annual sell turnover

$$\frac{\sum_t \text{sell}_t}{\bar{V} \times N/12}$$

$\bar{V}$ = average account value

Annual cost drag

$$\frac{\sum_t \text{Fee}_t}{\bar{V} \times N/12}$$